

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication

20250280371

Kind Code

A1

Publication Date

September 04, 2025

Inventor(s)

Loehr; Joachim et al.

REPORTING POWER HEADROOM INFORMATION

Abstract

Apparatuses, methods, and systems are disclosed. One apparatus is configured to determine that the UE is configured with a first subcarrier spacing (SCS) configuration for a first active bandwidth part (BWP) of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell. The apparatus is further configured to transmit a power headroom report (PHR) in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first physical uplink shared channel (PUSCH) on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

Inventors: Loehr; Joachim (Wiesbaden, DE), Golitschek Edler von Elbwart; Alexander Johann Maria (Darmstadt, DE), Bagheri; Hossein (Dearborn Heights, MI), Basu Mallick; Prateek (Dreieich, DE), Kuchibhotla; Ravi (Chicago, IL), Nangia; Vijay (Woodridge, IL)

Applicant: Lenovo (Singapore) Pte. Ltd. (Singapore, SG)

Family ID: 65995790

Appl. No.: 19/080695

Filed: March 14, 2025

Related U.S. Application Data

parent US continuation 18093520 20230105 parent-grant-document US 12279212 child US 19080695

parent US continuation 17461790 20210830 parent-grant-document US 11558828 child US 18093520

parent US continuation 16271685 20190208 parent-grant-document US 11109326 child US 17461790

us-provisional-application US 62628241 20180208

Publication Classification

Int. Cl.: H04W52/36 (20090101); H04W24/08 (20090101); H04W52/14 (20090101);
H04W52/34 (20090101); H04W72/0446 (20230101); H04W72/0453 (20230101)

U.S. Cl.:

CPC H04W52/365 (20130101); H04W24/08 (20130101); H04W52/34 (20130101);
H04W72/0446 (20130101); H04W72/0453 (20130101); H04W52/146 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims priority to U.S. patent application Ser. No. 18/093,520 entitled “REPORTING POWER HEADROOM INFORMATION” and filed on Jan. 5, 2023 for Joachim Loehr, Alexander Johann Maria Golitschek Edler von Elbwart, Hossein Bagheri, Prateek Basu Mallick, Ravi Kuchibhotla, and Vijay Nangia, which is incorporated herein by reference. U.S. application Ser. No. 18/093,520 claims priority to U.S. patent application Ser. No. 17/461,790 entitled “REPORTING POWER HEADROOM INFORMATION” and filed on Aug. 30, 2021 for Joachim Loehr, Alexander Johann Maria Golitschek Edler von Elbwart, Hossein Bagheri, Prateek Basu Mallick, Ravi Kuchibhotla, and Vijay Nangia, which is incorporated herein by reference. U.S. application Ser. No. 17/461,790 claims priority to U.S. patent application Ser. No. 16/271,685-now issued as U.S. Pat. No. 11,109,326-entitled “REPORTING POWER HEADROOM INFORMATION” and filed on Feb. 8, 2019 for Joachim Loehr, Alexander Johann Maria Golitschek Edler von Elbwart, Hossein Bagheri, Prateek Basu Mallick, Ravi Kuchibhotla, and Vijay Nangia, which is incorporated herein by reference. U.S. application Ser. No. 16/271,685 claims priority to U.S. Provisional Patent Application No. 62/628,241 entitled “PHR PROCEDURE WHEN AGGREGATING CARRIERS CONFIGURED WITH DIFFERENT TTI LENGTHS” and filed on Feb. 8, 2018 for Joachim Loehr, Alexander Johann Maria Golitschek Edler von Elbwart, Hossein Bagheri, Prateek Basu Mallick, Ravi Kuchibhotla, and Vijay Nangia, which is incorporated herein by reference.

FIELD

[0002] The subject matter disclosed herein relates generally to wireless communications and more particularly relates to reporting power headroom information.

BACKGROUND

[0003] In certain wireless communications networks, such as Long Term Evolution (LTE), a User Equipment (UE) reports extended power headroom report (PHR) or carrier aggregation, i.e., it reports power headroom (PH) info for each activated serving cell together with P.sub.CMAX, the total maximum UE transmit power. Because the subframe/TTI length is in LTE same for all carriers the PHR reporting subframes, the subframes to which the power headroom information refers are aligned. However, some wireless communication networks, such as the Third Generation Partnership Project (3GPP) Fifth Generation (5G) New Radio (NR), support carriers with different Orthogonal Frequency Division Multiplexing (OFDM) numerologies and/or different Transmission Time Intervals (TTIs).

BRIEF SUMMARY

[0004] Methods for reporting power headroom information are disclosed. Apparatuses and systems also perform the functions of the methods. The methods may also be embodied in one or more computer program products comprising executable code.

[0005] One method of a UE for reporting power headroom information includes being configured

with a first serving cell having a first TTI length and a second serving cell having a second TTI length, wherein the second TTI length is smaller than the first TTI length and having an uplink resource allocation for a first TTI on the first serving cell, wherein the first TTI overlaps in time with multiple second TTIs on the second serving cell. The third method includes calculating PH information for the second serving cell for a third TTI associated with a third TTI length that contains the first TTI on the first serving cell and transmitting the PH information in an uplink transmission on the first TTI on the first serving cell.

[0006] Another method of a UE may include determining that the UE is configured with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and transmitting a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0007] One method of a base station may include configuring a UE with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and receiving a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0008] A more particular description of the embodiments briefly described above will be rendered by reference to specific embodiments that are illustrated in the appended drawings. Understanding that these drawings depict only some embodiments and are not therefore to be considered to be limiting of scope, the embodiments will be described and explained with additional specificity and detail through the use of the accompanying drawings, in which:

[0009] FIG. 1 is a schematic block diagram illustrating one embodiment of a wireless communication system for reporting power headroom information;

[0010] FIG. 2 is a block diagram illustrating one embodiment of a Radio Access Network (RAN) for reporting power headroom information;

[0011] FIG. 3 is a block diagram illustrating another embodiment of a RAN for reporting power headroom information;

[0012] FIG. 4 is a schematic block diagram illustrating one embodiment of an apparatus that may be used for reporting power headroom information;

[0013] FIG. 5 is a block diagram illustrating a first embodiment of a scenario where a UE aggregates carriers configured with different transmission duration unit (TDU) lengths;

[0014] FIG. 6 is a block diagram illustrating a second embodiment of a scenario where a UE aggregates carriers configured with different TDU lengths;

[0015] FIG. 7 is a block diagram illustrating a third embodiment of a scenario where a UE aggregates carriers configured with different TDU lengths;

[0016] FIG. 8 is a block diagram illustrating a fourth embodiment of a scenario where a UE aggregates carriers configured with different TDU lengths;

[0017] FIG. 9 is a block diagram illustrating a fifth embodiment of a scenario where a UE aggregates carriers configured with different TDU lengths;

[0018] FIG. 10 is a block diagram illustrating a sixth embodiment of a scenario where a UE aggregates carriers configured with different TDU lengths;

[0019] FIG. **11** is a flow chart diagram illustrating a first method of reporting power headroom information;

[0020] FIG. **12** is a flow chart diagram illustrating a second method of reporting power headroom information; and

[0021] FIG. **13** is a flow chart diagram illustrating a third method of reporting power headroom information.

DETAILED DESCRIPTION

[0022] As will be appreciated by one skilled in the art, aspects of the embodiments may be embodied as a system, apparatus, method, or program product. Accordingly, embodiments may take the form of an entirely hardware embodiment, an entirely software embodiment (including firmware, resident software, micro-code, etc.) or an embodiment combining software and hardware aspects that may all generally be referred to herein as a “circuit,” “module” or “system.”

Furthermore, embodiments may take the form of a program product embodied in one or more computer readable storage devices storing machine readable code, computer readable code, and/or program code, referred hereafter as code. The storage devices may be tangible, non-transitory, and/or non-transmission. The storage devices may not embody signals. In a certain embodiment, the storage devices only employ signals for accessing code.

[0023] Any combination of one or more computer readable medium may be utilized. The computer readable medium may be a computer readable storage medium. The computer readable storage medium may be a storage device storing the code. The storage device may be, for example, but not limited to, an electronic, magnetic, optical, electromagnetic, infrared, holographic, micromechanical, or semiconductor system, apparatus, or device, or any suitable combination of the foregoing.

[0024] More specific examples (a non-exhaustive list) of the storage device would include the following: an electrical connection having one or more wires, a portable computer diskette, a hard disk, a random-access memory (RAM), a read-only memory (ROM), an erasable programmable read-only memory (EPROM or Flash memory), a portable compact disc read-only memory (CD-ROM), an optical storage device, a magnetic storage device, or any suitable combination of the foregoing. In the context of this document, a computer readable storage medium may be any tangible medium that can contain, or store a program for use by or in connection with an instruction execution system, apparatus, or device.

[0025] Code for carrying out operations for embodiments may be any number of lines and may be written in any combination of one or more programming languages including an object-oriented programming language such as Python, Ruby, Java, Smalltalk, C++, or the like, and conventional procedural programming languages, such as the “C” programming language, or the like, and/or machine languages such as assembly languages. The code may execute entirely on the user's computer, partly on the user's computer, as a stand-alone software package, partly on the user's computer and partly on a remote computer or entirely on the remote computer or server. In the latter scenario, the remote computer may be connected to the user's computer through any type of network, including a local area network (LAN) or a wide area network (WAN), or the connection may be made to an external computer (for example, through the Internet using an Internet Service Provider).

[0026] Reference throughout this specification to “one embodiment,” “an embodiment,” or similar language means that a particular feature, structure, or characteristic described in connection with the embodiment is included in at least one embodiment. Thus, appearances of the phrases “in one embodiment,” “in an embodiment,” and similar language throughout this specification may, but do not necessarily, all refer to the same embodiment, but mean “one or more but not all embodiments” unless expressly specified otherwise. The terms “including,” “comprising,” “having,” and variations thereof mean “including but not limited to,” unless expressly specified otherwise. An enumerated listing of items does not imply that any or all of the items are mutually exclusive, unless expressly

specified otherwise. The terms “a,” “an,” and “the” also refer to “one or more” unless expressly specified otherwise.

[0027] Furthermore, the described features, structures, or characteristics of the embodiments may be combined in any suitable manner. In the following description, numerous specific details are provided, such as examples of programming, software modules, user selections, network transactions, database queries, database structures, hardware modules, hardware circuits, hardware chips, etc., to provide a thorough understanding of embodiments. One skilled in the relevant art will recognize, however, that embodiments may be practiced without one or more of the specific details, or with other methods, components, materials, and so forth. In other instances, well-known structures, materials, or operations are not shown or described in detail to avoid obscuring aspects of an embodiment.

[0028] Aspects of the embodiments are described below with reference to schematic flowchart diagrams and/or schematic block diagrams of methods, apparatuses, systems, and program products according to embodiments. It will be understood that each block of the schematic flowchart diagrams and/or schematic block diagrams, and combinations of blocks in the schematic flowchart diagrams and/or schematic block diagrams, can be implemented by code. The code may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

[0029] The code may also be stored in a storage device that can direct a computer, other programmable data processing apparatus, or other devices to function in a particular manner, such that the instructions stored in the storage device produce an article of manufacture including instructions which implement the function/act specified in the schematic flowchart diagrams and/or schematic block diagrams block or blocks.

[0030] The code may also be loaded onto a computer, other programmable data processing apparatus, or other devices to cause a series of operational steps to be performed on the computer, other programmable apparatus, or other devices to produce a computer implemented process such that the code which execute on the computer or other programmable apparatus provide processes for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks.

[0031] The schematic flowchart diagrams and/or schematic block diagrams in the Figures illustrate the architecture, functionality, and operation of possible implementations of apparatuses, systems, methods, and program products according to various embodiments. In this regard, each block in the schematic flowchart diagrams and/or schematic block diagrams may represent a module, segment, or portion of code, which includes one or more executable instructions of the code for implementing the specified logical function(s).

[0032] It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the Figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. Other steps and methods may be conceived that are equivalent in function, logic, or effect to one or more blocks, or portions thereof, of the illustrated Figures.

[0033] Although various arrow types and line types may be employed in the flowchart and/or block diagrams, they are understood not to limit the scope of the corresponding embodiments. Indeed, some arrows or other connectors may be used to indicate only the logical flow of the depicted embodiment. For instance, an arrow may indicate a waiting or monitoring period of unspecified duration between enumerated steps of the depicted embodiment. It will also be noted that each block of the block diagrams and/or flowchart diagrams, and combinations of blocks in the block diagrams and/or flowchart diagrams, can be implemented by special purpose hardware-based systems that

perform the specified functions or acts, or combinations of special purpose hardware and code.

[0034] The description of elements in each figure may refer to elements of proceeding figures. Like numbers refer to like elements in all figures, including alternate embodiments of like elements.

[0035] Generally, the present disclosure describes systems, methods, and apparatus for reporting power headroom information when UE is configured with multiple uplink carriers, e.g., in a carrier aggregation deployment, respectively serving cells, for example by a UE communicating with a radio network using a first uplink (UL) carrier and a second UL carrier concurrently.

[0036] In various embodiments, the UE receives an UL resource allocation for a first transmission duration unit on the first UL carrier. Here, the first transmission duration unit overlaps in time with at least two second transmission duration units on the second UL carrier. The UE identifies a third transmission duration unit on the second UL carrier. Here, the third transmission duration unit comprises at least one of the second transmission duration units. The UE calculates PH information for the second UL carrier associated with the third transmission duration unit and reports the PH information in an UL transmission on the first transmission duration unit.

[0037] In some embodiments, each of the first and second UL carriers is associated with a different serving cell. In some embodiments, the UE may receive an UL resource allocation for at least one of the overlapping second transmission duration units on the second UL carrier.

[0038] In some embodiments, the first transmission duration unit corresponds to a slot on the first UL carrier and the second transmission duration unit corresponds to a slot on the second UL carrier. For example, the base unit may be a Next Generation (e.g., 5G) Node-B (gNB) in a 5G RAN, where the first UL carrier is configured with a first subcarrier spacing (SCS) and the second UL carrier is configured with a second SCS, the first SCS being smaller than the second SCS. Accordingly, the first UL carrier will have slots with longer time-duration than the second UL carrier, such that multiple second slots on the second UL carrier fully overlap with the first slot.

[0039] In such embodiments, the UE calculates PH information for a first physical uplink shared channel (PUSCH) scheduled on the first of the multiple second slots that fully overlaps with the first slot on the first UL carrier. Here, reporting PH information for the second UL carrier in an UL transmission on the first slot may include the UE transmitting on a PUSCH transmission a PHR that contains the PH information for the first PUSCH.

[0040] In some embodiments, the first transmission duration unit corresponds to a transmit time interval (TTI) of the first UL carrier and the second transmission duration unit corresponds to a TTI of the second UL carrier. In certain embodiments, the first UL carrier is configured with a first TTI length and the second UL carrier is configured with a second TTI length, wherein the first TTI length is larger than the second TTI length. For example, the base unit may be an LTE Evolved Node B (eNB), wherein the second TTI length corresponds to a shortened TTI (sTTI) length. Moreover, the first TTI length may also correspond to a sTTI.

[0041] In such embodiments, calculating PH information for the second UL carrier associated with the third transmission duration unit comprises the UE calculating PH for a TTI duration longer than the second TTI length. In certain embodiments, the length of the third transmission duration unit (e.g., third TTI) is greater than the length of the first TTI. In certain embodiments, the third transmission duration unit (e.g., third TTI) is equal to a subframe. In certain embodiments, calculating PH information for the second UL carrier associated with the third transmission duration unit comprises calculating PH information for a subframe containing the first TTI. In certain embodiments, the third transmission duration unit contains multiple TTIs of the second UL carrier (e.g., the third TTI overlaps multiple sTTIs on the second UL carrier).

[0042] In some embodiments, the UE calculates the PH information according to a predefined reference format. In certain embodiments, the processor calculates the PH information assuming the apparatus is not scheduled to transmit a PUSCH transmission in the third transmission duration unit.

[0043] In various embodiments, the reported PH information comprises a power headroom level computed based on the UL resource allocation. In some embodiments, TTIs of the second UL carrier

are configured with a smaller TTI length than TTIs of the first UL carrier, wherein the at least one second TTI on the second UL carrier has a shortened TTI length that is less than 1 millisecond. In some embodiments, the PH information is calculated based on an uplink resource allocation received for the third transmission duration unit. In such embodiments, the uplink transmission on the third transmission duration may be either stopped or dropped.

[0044] FIG. 1 depicts a wireless communication system **100** for receiving system information at a UE, according to embodiments of the disclosure. In one embodiment, the wireless communication system **100** includes at least one remote unit **105**, a RAN **120**, and a mobile core network **140**. The RAN **120** and the mobile core network **140** form a mobile communication network. The RAN **120** may be composed of a base unit **110** with which the remote unit **105** communicates using wireless communication links **115**. Even though a specific number of remote units **105**, base units **110**, wireless communication links **115**, RANs **120**, and mobile core networks **140** are depicted in FIG. 1, one of skill in the art will recognize that any number of remote units **105**, base units **110**, wireless communication links **115**, RANs **120**, and mobile core networks **140** may be included in the wireless communication system **100**.

[0045] In one implementation, the wireless communication system **100** is compliant with the 5G system specified in the 3GPP specifications. More generally, however, the wireless communication system **100** may implement some other open or proprietary communication network, for example, LTE or Worldwide Interoperability for Microwave Access (WiMAX), among other networks. The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

[0046] In one embodiment, the remote units **105** may include computing devices, such as desktop computers, laptop computers, personal digital assistants (PDAs), tablet computers, smart phones, smart televisions (e.g., televisions connected to the Internet), smart appliances (e.g., appliances connected to the Internet), set-top boxes, game consoles, security systems (including security cameras), vehicle on-board computers, network devices (e.g., routers, switches, modems), or the like. In some embodiments, the remote units **105** include wearable devices, such as smart watches, fitness bands, optical head-mounted displays, or the like. Moreover, the remote units **105** may be referred to as the UEs, subscriber units, mobiles, mobile stations, users, terminals, mobile terminals, fixed terminals, subscriber stations, user terminals, wireless transmit/receive unit (WTRU), a device, or by other terminology used in the art.

[0047] The remote units **105** may communicate directly with one or more of the base units **110** in the RAN **120** via UL and downlink (DL) communication signals. Furthermore, the UL and DL communication signals may be carried over the wireless communication links **115**. Here, the RAN **120** is an intermediate network that provides the remote units **105** with access to the mobile core network **140**.

[0048] In some embodiments, the remote units **105** communicate with an application server **151** via a network connection with the mobile core network **140**. For example, an application **107** (e.g., web browser, media client, telephone/VoIP application) in a remote unit **105** may trigger the remote unit **105** to establish a protocol data unit (PDU) session (or other data connection) with the mobile core network **140** via the RAN **120**. The mobile core network **140** then relays traffic between the remote unit **105** and the application server **151** in the packet data network **150** using the PDU session. Note that the remote unit **105** may establish one or more PDU sessions (or other data connections) with the mobile core network **140**. As such, the remote unit **105** may concurrently have at least one PDU session for communicating with the packet data network **150** and at least one PDU session for communicating with another data network (not shown).

[0049] The base units **110** may be distributed over a geographic region. In certain embodiments, a base unit **110** may also be referred to as an access terminal, an access point, a base, a base station, a Node-B, an eNB, a gNB, a Home Node-B, a relay node, or by any other terminology used in the art. The base units **110** are generally part of a RAN, such as the RAN **120**, that may include one or more

controllers communicably coupled to one or more corresponding base units **110**. These and other elements of radio access network are not illustrated but are well known generally by those having ordinary skill in the art. The base units **110** connect to the mobile core network **140** via the RAN **120**.

[0050] The base units **110** may serve a number of remote units **105** within a serving area, for example, a cell or a cell sector, via a wireless communication link **115**. The base units **110** may communicate directly with one or more of the remote units **105** via communication signals. Generally, the base units **110** transmit DL communication signals to serve the remote units **105** in the time, frequency, and/or spatial domain. Furthermore, the DL communication signals may be carried over the wireless communication links **115**. The wireless communication links **115** may be any suitable carrier in licensed or unlicensed radio spectrum. The wireless communication links **115** facilitate communication between one or more of the remote units **105** and/or one or more of the base units **110**.

[0051] In one embodiment, the mobile core network **140** is a 5G core network (5GC) or the evolved packet core network (EPC), which may be coupled to a packet data network **150**, like the Internet and private data networks, among other data networks. A remote unit **105** may have a subscription or other account with the mobile core network **140**. Each mobile core network **140** belongs to a single public land mobile network (PLMN). The present disclosure is not intended to be limited to the implementation of any particular wireless communication system architecture or protocol.

[0052] The mobile core network **140** includes several network functions (NFs). As depicted, the mobile core network **140** includes multiple user plane functions (UPFs) **145**. The mobile core network **140** also includes multiple control plane functions including, but not limited to, an access and mobility management function (AMF) **141** that serves the RAN **120**, a session management function (SMF) **143**, and a policy control function (PCF) **147**. In certain embodiments, the mobile core network **140** may also include an authentication server function (AUSF), a unified data management function (UDM) **149**, a network repository function (NRF) (used by the various NFs to discover and communicate with each other over APIs), or other NFs defined for the 5GC.

[0053] Although specific numbers and types of network functions are depicted in FIG. **1**, one of skill in the art will recognize that any number and type of network functions may be included in the mobile core network **140**. Moreover, where the mobile core network **140** is an EPC, the depicted network functions may be replaced with appropriate EPC entities, such as a mobility management entity (MME), serving gateway (S-GW), packet gateway (P-GW), home subscriber server (HSS), and the like. In certain embodiments, the mobile core network **140** may include a AAA server.

[0054] In various embodiments, the mobile core network **140** supports different types of mobile data connections and different types of network slices, wherein each mobile data connection utilizes a specific network slice. Here, a “network slice” refers to a portion of the mobile core network **140** optimized for a certain traffic type or communication service. In certain embodiments, the various network slices may include separate instances of network functions, such as the SMF **143** and UPF **145**. In some embodiments, the different network slices may share some common network functions, such as the AMF **141**. The different network slices are not shown in FIG. **1** for ease of illustration, but their support is assumed.

[0055] While FIG. **1** depicts components of a 5G RAN and a 5G core network, the described embodiments for PHR reporting **125** in a wideband carrier apply to other types of communication networks, including IEEE 802.11 variants, universal mobile telecommunications system (UMTS), LTE variants, CDMA2000, Bluetooth, and the like. For example, in an LTE/EPC variant, the AMF **141** may be mapped to an MME, the SMF **143** may be mapped to a control plane portion of a P-GW, the UPF **145** may be mapped to a S-GW and a user plane portion of the P-GW, etc.

[0056] To assist the base unit **110** to schedule uplink transmission resources to different remote units **105** in an appropriate way, each remote unit **105** reports its available PH to the base unit **110**, e.g., using a power headroom report (PHR) **125**. Using a received PHR, the base unit **110** may determine

how much more uplink bandwidth per sub-frame a remote unit **105** is capable of using, i.e., how close to its transmission power limits the remote unit **105** is operating. The PH indicates the difference between the maximum UE uplink transmit power and the estimated power for UL-SCH transmission. In various embodiments, the remote unit **105** power headroom (PH) in dB valid for sub-frame “i” is defined by:

[00001]

$$PH(i) = P_{C_{MAX}} - \{10 \log_{10}(M_{PUSCH}(i)) + P_{0,PUSCH}(j) + (j) \cdot \text{Math. PL} + T_F(i) + f(i)\} \quad \text{Equation 1}$$

[0057] Here, P.sub.CMAX is the total maximum UE transmit power and is a value chosen by the user equipment in the given range of P.sub.CMAX_L and P.sub.CMAX_H based on the following constraints:

$$P_{C_{MAX_L}} \leq P_{C_{MAX}} \leq P_{C_{MAX_H}} \quad \text{Equation 2}$$

$$P_{C_{MAX_L}} = \min(P_{EMAX} - T_C, P_{PowerClass} - MPR - AMPR - T_C) \quad \text{Equation 3}$$

$$P_{C_{MAX_H}} = \min(P_{EMAX}, P_{PowerClass}) \quad \text{Equation 4}$$

[0058] Here, P.sub.EMAX is a value signaled by the network. The MPR is a power reduction value used to control the adjacent channel leakage ratio (ACLR) associated with the various modulation schemes and the transmission bandwidth. AMPR is the additional maximum power reduction. It is a band specific value and applied by the UE when configured by the network. One example of values for ΔT.sub.C, MPR and AMPR may be found in 3GPP TS36.101.

[0059] In various embodiments, the remote unit **105** sends the PHR **125** as a medium access control-control element (MAC-CE). The base unit **110** may configure parameters to control various triggers for reporting power headroom depending on the system load and the requirements of its scheduling algorithm.

[0060] In various embodiments, the range of the power headroom report is from +40 to −23 dB. Note that negative part of the range enables the remote unit **105** to signal to the base unit **110** the extent to which it has received an UL grant which would require more transmission power than the remote unit **105** has available. The base unit **110** may then reduce the amount of uplink resources in a subsequent grant (dynamic or semi-static), thus freeing up transmission resources which could be then allocated to other remote units.

[0061] In various embodiments, the power headroom report **125**, e.g., PHR MAC-CE, may only be sent in a sub-frame for which the remote unit **105** has a valid uplink resource, i.e., a PUSCH resource. In general, the PHR **125** relates to the sub-frame in which it is sent and is therefore an estimation or prediction rather than a direct measurement (because the remote unit **105** cannot directly measure its actual transmission power headroom for the subframe in which the report is to be transmitted).

[0062] In various embodiments, the remote unit **105** is configured for carrier aggregation, wherein the remote unit **105** used at least a first carrier and a second carrier concurrently. Each component carrier (e.g., the first and second carriers) may be associated with a different serving cell. Where the remote unit **105** is configured with multiple concurrent serving cells, the power headroom defined in Equation 1 is calculated and reported for each serving cell/component carrier. For the case of carrier aggregation, the remote unit **105** must consider both the total maximum UE transmit power P.sub.CMAX and a component carrier-specific maximum transmit power P.sub.CMAX,c.

[0063] Because simultaneous PUSCH and physical uplink control channel (PUCCH) transmission is supported in carrier aggregation, two different types of PH types are supported for CA. PH type 1 indicates the difference between P.sub.CMAX,c and estimated PUSCH power, while PH type 2 indicates the difference between P.sub.CMAX,c and estimated power of PUSCH and PUCCH combined. Note that PH type 2 is only applicable for primary cell (PCell), whereas PH type 1 is applicable for both PCell and secondary cell (SCell). Because it is beneficial for the base unit **110** to always know the power situation for all activated uplink carrier/serving carrier for future uplink scheduling, the remote unit **105** may transmit an extended PH MAC-CE on one of the serving cells

(PCell and one or more SCells) which has a valid uplink resource for PUSCH. The extended PH MAC-CE includes power headroom information (Type1/Type2) for each activated uplink component carrier.

[0064] While the following solutions are discussed in the context of carrier aggregation, the principles described herein are also applicable to Dual connectivity (DC) which allows a remote unit **105** to receive data simultaneously from different base units **110** in order to boost the performance in a heterogeneous network with dedicated carrier deployment. In Dual Connectivity when a PHR has been triggered, the UE sends power headroom information for all activated cells (including serving cells of both cell groups) to the eNB. When UE reports PH info of secondary cell group (SCG) cells to the main base unit **110** (e.g., main eNB (MeNB)) or PH info of main/master cell group (MCG) cells to the secondary base unit **110** (e.g., secondary eNB (SeNB)), Type2 PH information for the PUCCH cell (PUCCH for the SCG) is included. Power headroom info for the serving cells in the other CG may be calculated based on some reference format (e.g., virtual PHR) or based on actual PUSCH/PUCCH transmissions.

[0065] In various embodiments, the remote unit **105** may be configured with a Short Processing Time (SPT) and a shorter TTI length. Short transmission time interval (Short TTI) provides support for TTI length shorter than 1 ms DL-SCH and UL-SCH. To support the short TTI, the associated control channels, shortened physical downlink control channel (sPDCCH), (i.e., containing downlink control information (DCI) for short TTI operation, referred to as “sDCI”) and shortened physical uplink control channel (sPUCCH) are also transmitted with duration shorter than 1 ms. Over the physical layer, DL and UL transmissions use either slots or subslots when short TTI is configured. Recall that in LTE there are 2 slots of 7 OFDM (or single carrier frequency division multiple access (SC-FDMA)) symbol duration in a subframe. As used herein, a “subslot” refers to a transmission duration unit of either 2 OFDM/SC-FDMA symbol or 3 OFDM/SC-FDMA symbol duration. As such, three “subslots” fit within a slot. To support the short TTI, the remote unit **105** transmits slot-based (or subslot-based) PUSCH (also referred to as “shortened PUSCH” or “sPUSCH”).

[0066] In various embodiments, the RAN **120** may support different OFDM numerologies, i.e., sub-carrier spacing (SCS), cyclic prefix (CP) length, in a single framework, e.g., to support use cases/deployment scenarios with diverse requirements in terms of data rates, latency, and coverage. For example, enhanced mobile broadband (eMBB) is to support peak data rates (e.g., 20 Gbps for downlink and 10 Gbps for uplink) while ultra-reliability and low-latency communications (URLLC) is to support ultra-low latency (0.5 ms for UL and DL each for user plane latency) and high reliability (1-10⁻⁵ within 1 ms). Therefore, the OFDM numerology that is suitable for one use case might not work well for another. Note that different OFDM numerologies have different SCSs, affecting OFDM symbol duration, CP duration, and number of symbols per scheduling interval. Different numerologies may occur across different carrier(s) for a given UE as well as different numerologies within the same carrier for a given UE, i.e., different OFDM numerologies are multiplexed in frequency-domain and/or time-domain within the same carrier or across different carriers.

[0067] When carrier aggregation is combined with different numerologies for NR (e.g., 5G radio access) or, for LTE, with shortened TTI, one TDU (e.g., NR slot or LTE TTI) of a carrier can overlap (coincide) with multiple TDUs of another carrier. In this case the base unit **110**, may not be aware which TDU the power headroom information refers to. For example, in a scenario where an extended PHR report is triggered and subsequently transmitted in a slot/TTI, which overlaps with multiple slots/TTIs on a different carrier, the base unit **110** does not know which of the overlapped slot/TTI from the multiple slots/TTIs is the reference for the PH calculation. Without knowledge of the reference TDU, the base unit **110** may establish its future scheduling decisions on wrong assumptions, i.e., how close the UE is operating on the power limit, which may lead to either power scaling or under-utilization of resources. Note that data transmissions may be scheduled to span one

or multiple UL TDUs (e.g., slots/TTIs). Similarly, multiple data transmissions (e.g., PUSCH transmissions) also may be scheduled within one slot, which is also referred to as, e.g., sub-slot as outlined below.

[0068] In various embodiments, the remote unit **105** is configured with different UL TDU lengths for different serving cells, e.g., for a carrier aggregation. With possible simultaneous UL transmissions using different PUSCH durations across serving cells, a base unit **110**, e.g., gNB or eNB, needs to know which TDU a power headroom calculation is based on, so that it correctly interprets a received PHR and to enable the base unit **110** to schedule subsequent transmissions (e.g., sTTI/TTI/slot) properly. As discussed above, the PHR provides the base unit **110** with information on path loss, TPC status and the used MPR for the corresponding uplink transmission.

[0069] In certain embodiments, the remote unit **105** may receive indications from the base unit **110** of sets of frequency domain resource blocks for possible PUSCH data transmission in uplink of a first TDU length on a first component carrier and a second TDU length on the second component carrier. In one embodiment, the remote unit **105**, having been allocated resources for PUSCH on the first carrier, then identifies a TDU on the second carrier computes a PHR for the second carrier based on the identified TDU. In certain embodiments, the remote unit **105** may derive (and indicate to the base unit **110**) a reference TDU index (e.g., slot index or TTI index) associated with the PHR for the second carrier. The remote unit **105** transmits the PHR for the second carrier to the base unit **110**.

[0070] FIG. 2 depicts an access network **200** for reporting power headroom information, according to embodiments of the disclosure. The access network **200** include a UE **205** that uses two UL carriers concurrently, a first component carrier **220** (denoted “CC1”) and a second component carrier **230** (denoted “CC2”). In various embodiments, each component carrier is associated with a different serving cell. In the depicted embodiment, the first component carrier **220** and the second component carrier **230** are associated with the same RAN node **210** (denoted “gNB-1”), e.g., a Carrier Aggregation scenario. Here, the RAN node **210** may be gNB in a 5G-RAN.

[0071] The first component carrier **220** has transmission duration units of longer length than the second component carrier **230**. In various embodiments, these transmission duration units correspond to slots in NR with the first component carrier being configured with a smaller SCS than the second component carrier. As such, multiple slots on the second component carrier **230** fit within a single slot on the first component carrier **220**. In the depicted embodiment, one slot of the first component carrier **220** overlaps with two slots of the second component carrier **230**.

[0072] As depicted, the first component carrier **220** includes a first slot **221** (denoted “slot-0”), a second slot **222** (denoted “slot-1”), a third slot **223** (denoted “slot-2”), and a fourth slot **224** (denoted “slot-3”). At the same time, the second component carrier **230** includes a first slot **231** (denoted “slot-0”), a second slot **232** (denoted “slot-1”), a third slot **233** (denoted “slot-2”), a fourth slot **234** (denoted “slot-3”), a fifth slot **235** (denoted “slot-4”), a sixth slot **236** (denoted “slot-5”), a seventh slot **237** (denoted “slot-6”), and an eighth slot **238** (denoted “slot-7”). While the depicted embodiment shows alignment of the slot boundaries, in other embodiments the slot boundaries of the first component carrier **220** do not coincide with slot boundaries of the second component carrier **230**.

[0073] In the access network **200**, the UE **205** is allocated uplink resources in the slot **223**, denoted “slot-2”, on the first component carrier **220**. Here, the allocated resources may correspond to an active bandwidth part (BWP) on the first component carrier **220**. Note that the slot **223** overlaps with the slots **235** and **236** on the second component carrier **230**. Here, the UE **205** provides a type1 PHR for slot **223** of the first component carrier **220** in a PUSCH transmission on slot **223**.

Moreover, the UE **205** provides a PHR for the first PUSCH, if any, on the first of the multiple slots on the second component carrier **230** (e.g., on an active BWP of the second component carrier **230**) that fully overlaps with the slot **223**. Here, the slot **235** is the first slot on the second component carrier **230** to fully overlap with the slot **223**. Thus, the UE **205** calculates a PHR (e.g., a type1 PHR)

for the slot **235** and sends it to the network in the PUSCH transmission on slot **223** (e.g., sends the PHR **240** over the first component carrier **220**).

[0074] In certain embodiments, the slot **235** and slot **236** are scheduled for uplink transmission (e.g., PUSCH). Here, the uplink grants for these slots may include dynamic grants and/or configured (e.g., semi-persistent) grants. In one embodiment, the slots **235** and **236** may be scheduled individually. In another embodiment, the slots **235** and **236** are scheduled via a multi-slot grant. In certain embodiments, the UE **205** computes an “actual” PHR for the second component carrier. As used herein, “actual” PH refers to a power headroom level calculated based on actual transmissions. In contrast, “virtual” PH refers to a power headroom level calculated based on a predefined reference format.

[0075] FIG. **3** depicts an access network **300** for reporting power headroom information, according to embodiments of the disclosure. The access network **300** includes the UE **205** that uses two UL carriers concurrently, a first component carrier **320** (denoted “CC1”) and a second component carrier **330** (denoted “CC2”). In the depicted embodiment, the first component carrier **320** and the second component carrier **330** are associated with a RAN node **310** (denoted “eNB-1”). Here, the RAN node **310** may be eNB in an LTE-RAN.

[0076] The first component carrier **320** has transmission duration units of longer length than the second component carrier **330**. In various embodiments, these transmission duration units correspond to TTIs, for example short TTIs (sTTIs) in LTE as depicted, with the first component carrier being configured with a longer TTI length than the second component carrier. As such, multiple sTTIs on the second component carrier **330** fit within a single sTTI on the first component carrier **320**. In the depicted embodiment, one sTTI of the first component carrier **320** overlaps with three sTTIs of the second component carrier **330**. Here, the first component carrier is configured with a slot-length TTI (e.g., 7 OFDM symbols in duration) and the second component carrier **330** is configured with subslot-length TTI (e.g., 2 or 3 OFDM symbols in duration; here in a “2-2-3” pattern, such that every three sTTIs add up to 7 OFDM symbols).

[0077] As depicted, the first component carrier **320** includes a first sTTI **321** (denoted “sTTI-1”), a second sTTI **322** (denoted “sTTI-2”), a third sTTI **323** (denoted “sTTI-3”), and a fourth sTTI **324** (denoted “sTTI-4”). At the same time, the second component carrier **330** includes a first sTTI **331** (denoted “sTTI-1”), a second sTTI **332** (denoted “sTTI-2”), a third sTTI **333** (denoted “sTTI-3”), a fourth sTTI **334** (denoted “sTTI-4”), a fifth sTTI **335** (denoted “sTTI-5”), a sixth sTTI **336** (denoted “sTTI-6”), a seventh sTTI **337** (denoted “sTTI-7”), an eighth sTTI **338** (denoted “sTTI-8”), a ninth sTTI **339** (denoted “sTTI-9”), a tenth sTTI **340** (denoted “sTTI-10”), an eleventh sTTI **341** (denoted “sTTI-11”), and a twelfth sTTI **342** (denoted “sTTI-12”). While the depicted embodiment shows alignment of the sTTI boundaries, in other embodiments the sTTI boundaries of the first component carrier **320** do not coincide with sTTI boundaries of the second component carrier **330**.

[0078] In the access network **300**, the UE **205** is allocated uplink resources in the sTTI **323**, denoted “sTTI-3”, on the first component carrier **320**. Note that the sTTI **323** overlaps with the sTTIs **337**, **338** and **339** on the second component carrier **330**. Here, the UE **205** provides a PHR for sTTI **323** of the first component carrier **320** in a PUSCH transmission on sTTI **323**. Moreover, the UE **205** provides a PHR **350** for time duration unit on the second component carrier **330** that is larger than the sTTI length(s) of the second component carrier **330**. In the depicted embodiment, the UE **205** calculates a PHR for a subframe-length TDU on the second component carrier **330**. Here, the UE **205** computes the Power headroom for the subframe **345** on the second component carrier **330** containing the sTTI **323** with the PUSCH allocation on the first component carrier **320**. In other embodiments, the UE **205** calculates a PHR for a slot-length TDU on the second component carrier **330**, denoted as “TTI-3” **347**.

[0079] In certain embodiments, the sTTIs **337-339** are scheduled for uplink transmission (e.g., PUSCH). Here, the uplink grants for these sTTIs may include dynamic grants and/or configured (e.g., semi-persistent) grants. In one embodiment, the sTTIs **337-339** may be scheduled individually.

In another embodiment, the sTTIs 337-339 are scheduled via a multi-sTTI grant. In certain embodiments, the UE 205 computes an “virtual” PHR for the second component carrier. As used herein, “virtual” PH refers to a power headroom level calculated based on a predefined reference format. In other embodiments, the UE 205 may compute an “actual” PHR for the second component carrier.

[0080] FIG. 4 depicts a user equipment apparatus 400 that may be used for reporting power headroom information, according to embodiments of the disclosure. The user equipment apparatus 400 may be one embodiment of the remote unit 105 and/or the UE 205, described above.

Furthermore, the user equipment apparatus 400 may include a processor 405, a memory 410, an input device 415, an output device 420, a transceiver 425 for communicating with one or more base units 110.

[0081] As depicted, the transceiver 425 may include a transmitter 430 and a receiver 435. The transceiver 425 may also support one or more network interfaces 440, such as the Uu interface used to communicate with a gNB, or other suitable interface for communicating with the RAN 120. In some embodiments, the input device 415 and the output device 420 are combined into a single device, such as a touchscreen. In certain embodiments, the user equipment apparatus 400 may not include any input device 415 and/or output device 420.

[0082] In various embodiments, the processor 405 receives (i.e., via the transceiver 425) a service configuration for a first serving cell for PUSCH transmissions having a first SCS and a second serving cell for PUSCH transmissions having a second SCS, where the first SCS is smaller than the second SCS. The processor 405 receives (i.e., via the transceiver 425) an uplink resource allocation for a first slot on the first serving cell, where the first slot overlaps in time with multiple second slots on the second serving cell. Additionally, the processor receives (i.e., via the transceiver 425) an uplink resource allocation for at least one of the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. The processor 405 calculates PH information for the second serving cell for a first PUSCH scheduled on a first slot of the multiple second slots that fully overlaps with the first slot on the first serving cell and controls the transceiver 425 to transmit the PH information in an uplink transmission on the first slot on the first serving cell.

[0083] In some embodiments, the processor 405 further receives (via the transceiver 425) a second uplink resource allocation for at least one of the overlapping second slots on the second uplink carrier. In some embodiments, transmitting the PH information in an uplink transmission on the first slot on the first serving cell includes transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH scheduled on the first of the multiple second slots that is fully overlapped by the first slot.

[0084] In some embodiments, transmitting the PH information includes transmitting a Type 1 power headroom report. In some embodiments, transmitting the PH information includes transmitting a PHR MAC-CE in the uplink transmission in an allocated uplink allocation on the first serving cell.

[0085] In some embodiments, the first slot corresponds to a TTI of the first serving cell and the second slots correspond to TTIs of the second serving cell. In certain embodiments, the PH information for the second serving cell is calculated for TTI duration that contains multiple TTIs of the second serving cell.

[0086] In some embodiments, the first serving cell is configured with a first TTI length and the second serving cell is configured with a second TTI length, where the first TTI length is larger than the second TTI length. In certain embodiments, calculating PH information for the second serving cell includes calculating PH information for a subframe that uses the first TTI length.

[0087] In certain embodiments, calculating PH information for the second serving cell includes calculating PH for a TTI duration longer than the second TTI length. In one embodiment, the PH for the second serving cell is calculated for TTI duration that is greater than the length of the first TTI. In another embodiment, the PH for the second serving cell is calculated for TTI duration that is equal to a subframe.

[0088] In some embodiments, the PH information is calculated according to a predefined reference format. In certain embodiments, the PH information is calculated assuming the UE is not scheduled to transmit a PUSCH in the at least two second slots that fully overlaps with the first slot.

[0089] In some embodiments, the at least one of the second slots on the second serving cell has a shortened TTI length that is less than 1 millisecond. In some embodiments, the reported PH information includes a power headroom level computed based on the uplink resource allocation.

[0090] In some embodiments, the PH information is calculated based on an uplink resource allocation received for a transmission duration corresponding to the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. In certain embodiments, an uplink transmission on the transmission duration is one of: stopped and dropped.

[0091] In various embodiments, the transceiver **425** communicates with a base unit using a first uplink carrier and a second uplink carrier concurrently. Here, each carrier has a different transmission duration unit length, wherein the first uplink carrier has a longer transmission duration unit length than the second uplink carrier. At some point in time, the processor **405** receives an uplink resource allocation for a first transmission duration unit on the first uplink carrier and determines a third transmission duration unit on the second uplink carrier. Here, the first transmission duration unit overlaps in time with at least two second transmission duration units on the second uplink carrier and the third transmission duration unit comprises at least one of the second transmission duration units. The processor **405** calculates PH information for the second uplink carrier associated with the third transmission duration unit and reports, via the transceiver **425**, the PH information in an uplink transmission on the first transmission duration unit.

[0092] In some embodiments, each of the first and second uplink carriers is associated with a different serving cell. In some embodiments, the processor further receives an uplink resource allocation for at least one of the overlapping second transmission duration units on the second uplink carrier.

[0093] In some embodiments, the first transmission duration unit corresponds to a slot on the first uplink carrier and the second transmission duration unit corresponds to a slot on the second uplink carrier. In certain embodiments, multiple second slots on the second uplink carrier fully overlap with the first slot. In such embodiments, the processor **405** calculates PH information for a first PUSCH scheduled on the first of the multiple second slots that fully overlaps with the first slot on the first uplink carrier, wherein reporting PH information for the second uplink carrier in an uplink transmission on the first slot comprises transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH. In certain embodiments, the first uplink carrier is configured with a first SCS and the second uplink carrier is configured with a second SCS, wherein the first SCS is smaller than the second SCS.

[0094] In some embodiments, the first transmission duration unit corresponds to a TTI of the first uplink carrier and the second transmission duration unit corresponds to a TTI of the second uplink carrier. In certain embodiments, the first uplink carrier is configured with a first TTI length and the second uplink carrier is configured with a second TTI length, wherein the first TTI length is larger than the second TTI length. In such embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit comprises calculating PH for a TTI duration longer than the second TTI length.

[0095] In certain embodiments, the length of the third transmission duration unit is greater than the length of the first TTI. In certain embodiments, the third transmission duration unit is equal to a subframe. In certain embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit comprises calculating PH information for a subframe containing the first TTI. In certain embodiments, the third transmission duration unit contains multiple TTIs of the second uplink carrier.

[0096] In some embodiments, the processor **420** calculates the PH information according to a predefined reference format. In certain embodiments, the processor **420** calculates the PH

information assuming the apparatus is not scheduled to transmit a PUSCH in the third transmission duration unit.

[0097] In various embodiments, the reported PH information comprises a power headroom level computed based on the uplink resource allocation. In some embodiments, the at least one of the second transmission duration units on the second uplink carrier (e.g., second TTI on the second uplink carrier) has a shortened TTI length that is less than 1 millisecond. In some embodiments, the PH information is calculated based on an uplink resource allocation received for the third transmission duration unit. In such embodiments, the uplink transmission on the third transmission duration may be either stopped or dropped.

[0098] In some embodiments, the transceiver **425** receives a first indication (e.g., first UL resource grant) from a mobile communication network (e.g., from the base unit **110**) indicating a first set of frequency domain resource blocks for possible PUSCH data transmission and at least one UL TDU of a first TDU length on a first component carrier. The transceiver **425** may also receive a second indication (e.g., second UL resource grant) from the mobile communication network indicating a second set of frequency domain resource blocks for possible PUSCH data transmission and at least one uplink TDU of a second TDU length on the second component carrier. Here, the first component carrier and the second component carrier are configured with different TDU lengths (e.g., different TTI/sTTI lengths in an LTE deployment or different slot lengths in an NR deployment).

[0099] The processor **405** computes a first PHR based on at least one of: transmissions of the first TDU length only being present in a first TDU on the first carrier, and a reference format. In certain embodiments, the processor **405** may also select at least one second TDU of the second TDU length on the second component carrier and compute a second PHR based on at least one of: transmissions of the second TDU length only being present in the selected second TDU on the second component carrier, and a reference format. Moreover, the processor **405** may derive a reference TDU index associated with the second PHR.

[0100] The processor **405** controls the transceiver **425** to transmit at least the first PHR and the second PHR to the base unit (e.g., gNB or eNB). Here, the second TDU length is shorter than the first TDU length, such that the first TDU encompasses one or more TDUs of the second TDU length. For example, an integer number of TDUs of the second TTI length may fit within the time duration of the first TDU.

[0101] In one embodiment, the second indication is a radio resource control (RRC) configuration assigning the second set of frequency domain resource blocks for possible PUSCH data transmissions (e.g., a configured UL grant, such as semi-persistent scheduling). Additionally, the RRC configuration may further indicate at least one of the modulation and coding scheme (MCS) and a transport block size (TBS) index. In such embodiments, the second PHR may contain a power headroom value computed based on transmissions of the second TTI length only being present in the second TTI on the second component carrier.

[0102] Additionally, in the above embodiments computing the second PHR may be based on the allocation of the second set of frequency domain resource blocks, irrespective of the presence of transmissions in the second TTI on the second component carrier (e.g., the processor **405** calculates the second PHR assuming that PUSCH transmissions will occur on the one or more second TDUs, even if PUSCH is actually not transmitted on the one or more second TDUs). In such embodiments, a configured maximum transmit power value (e.g., $P_{\text{sub.CMAX},c}$) may be included in the second PHR.

[0103] In certain embodiments, the first PHR and at least the second PHR are transmitted via a single PHR MAC-CE. In certain embodiments, the reference TDU index is derived based on at least the second TDU index within the first TDU.

[0104] In some embodiments, the processor **405** selects a second TDU from a set of TDUs of the second TDU length within the first TDU. Here, the processor **405** may select the earliest scheduled

TDU of the second TDU length within the first TDU for uplink transmission. In another embodiment, the processor **405** may select the second TDU from the set of TDUs of the second TDU length within the first TDU based on the scheduled TDU of the second TDU length within the first TDU for uplink transmission having a smallest power headroom field value among the scheduled TDU of the second TDU length within the first TDU for uplink transmission. In a third embodiment, the second TDU selected from a set of TDUs of the second TDU length within the first TDU based on the earliest scheduled TDU of the second TDU length within the first TDU for uplink transmission if no PHR is due on any earlier TTI of the second TTI length within the first TDU, and otherwise based on the earliest TDU of the second TDU length within the first TDU for which a PHR is due.

[0105] In some embodiments, computing the second PHR is based on the reference format (e.g., a virtual PHR). In such embodiments, the second PHR may be computed according to a third TTI with a third TTI length. In one such embodiment, the third TTI length is equal to one subframe (e.g., 1 ms). In an alternate embodiment, the third TTI length is the same as the first TTI length. Also, in such embodiments, uplink transmissions of the third TTI length may not be configured for the second component carrier. Moreover, in these embodiments the transmit power control command value may be a fixed value. Further, the second PHR calculation may be associated with fixed resource block allocation and a transmit power control command value the fixed value of the transmit power control command may be selected based on the third TTI length. In certain embodiments, the first component carrier belongs to the first PUCCH group and the second component carrier belongs to a second PUCCH group.

[0106] The memory **410**, in one embodiment, is a computer readable storage medium. In some embodiments, the memory **410** includes volatile computer storage media. For example, the memory **410** may include a RAM, including dynamic RAM (DRAM), synchronous dynamic RAM (SDRAM), and/or static RAM (SRAM). In some embodiments, the memory **410** includes non-volatile computer storage media. For example, the memory **410** may include a hard disk drive, a flash memory, or any other suitable non-volatile computer storage device. In some embodiments, the memory **410** includes both volatile and non-volatile computer storage media.

[0107] In some embodiments, the memory **410** stores data related to reporting power headroom information. For example, the memory **410** may store one or more power headroom reports, e.g., of the first and second types described herein. Additionally, the memory **410** may store data for reporting power headroom information, such as PH values, resource allocations, TTI index, and the like. In certain embodiments, the memory **410** also stores program code and related data, such as an operating system or other controller algorithms operating on the remote unit **105**.

[0108] The input device **415**, in one embodiment, may include any known computer input device including a touch panel, a button, a keyboard, a stylus, a microphone, or the like. In some embodiments, the input device **415** may be integrated with the output device **420**, for example, as a touchscreen or similar touch-sensitive display. In some embodiments, the input device **415** includes a touchscreen such that text may be input using a virtual keyboard displayed on the touchscreen and/or by handwriting on the touchscreen. In some embodiments, the input device **415** includes two or more different devices, such as a keyboard and a touch panel.

[0109] The output device **420**, in one embodiment, is designed to output visual, audible, and/or haptic signals. In some embodiments, the output device **420** includes an electronically controllable display or display device capable of outputting visual data to a user. For example, the output device **420** may include, but is not limited to, an LCD display, an LED display, an OLED display, a projector, or similar display device capable of outputting images, text, or the like to a user. As another, non-limiting, example, the output device **420** may include a wearable display separate from, but communicatively coupled to, the rest of the user equipment apparatus **400**, such as a smart watch, smart glasses, a heads-up display, or the like. Further, the output device **420** may be a component of a smart phone, a personal digital assistant, a television, a table computer, a notebook

(laptop) computer, a personal computer, a vehicle dashboard, or the like.

[0110] In certain embodiments, the output device **420** includes one or more speakers for producing sound. For example, the output device **420** may produce an audible alert or notification (e.g., a beep or chime). In some embodiments, the output device **420** includes one or more haptic devices for producing vibrations, motion, or other haptic feedback. In some embodiments, all or portions of the output device **420** may be integrated with the input device **415**. For example, the input device **415** and output device **420** may form a touchscreen or similar touch-sensitive display. In other embodiments, the output device **420** may be located near the input device **415**.

[0111] As discussed above, the transceiver **425** communicates with one or more network functions of a mobile communication network via one or more access networks. The transceiver **425** operates under the control of the processor **405** to transmit messages, data, and other signals and also to receive messages, data, and other signals. For example, the processor **405** may selectively activate the transceiver **425** (or portions thereof) at particular times in order to send and receive messages.

[0112] In various embodiments, the transceiver **425** includes at least one transmitter **430** and at least one receiver **435**. One or more transmitters **430** may be used to provide UL communication signals to a base unit **110**, such as the PUSCH transmissions containing PHR described herein. Similarly, one or more receivers **435** may be used to receive DL communication signals from the base unit **110**, as described herein. Although only one transmitter **430** and one receiver **435** are illustrated, the user equipment apparatus **400** may have any suitable number of transmitters **430** and receivers **435**. Further, the transmitter(s) **425** and the receiver(s) **430** may be any suitable type of transmitters and receivers. In one embodiment, the transceiver **425** includes a first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and a second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum.

[0113] In certain embodiments, the first transmitter/receiver pair used to communicate with a mobile communication network over licensed radio spectrum and the second transmitter/receiver pair used to communicate with a mobile communication network over unlicensed radio spectrum may be combined into a single transceiver unit, for example a single chip performing functions for use with both licensed and unlicensed radio spectrum. In some embodiments, the first transmitter/receiver pair and the second transmitter/receiver pair may share one or more hardware components. For example, certain transceivers **425**, transmitters **430**, and receivers **435** may be implemented as physically separate components that access a shared hardware resource and/or software resource, such as for example, the network interface **440**.

[0114] In various embodiments, one or more transmitters **430** and/or one or more receivers **435** may be implemented and/or integrated into a single hardware component, such as a multi-transceiver chip, a system-on-a-chip, an application specific integrated circuit (ASIC), or other type of hardware component. In certain embodiments, one or more transmitters **430** and/or one or more receivers **435** may be implemented and/or integrated into a multi-chip module. In some embodiments, other components such as the network interface **440** or other hardware components/circuits may be integrated with any number of transmitters **430** and/or receivers **435** into a single chip. In such embodiment, the transmitters **430** and receivers **435** may be logically configured as a transceiver **425** that uses one more common control signals or as modular transmitters **430** and receivers **435** implemented in the same hardware chip or in a multi-chip module.

[0115] FIG. 5 depicts a first scenario **500** where the UE **205** aggregates serving cells (e.g., in a carrier aggregation deployment) with different TDU lengths, according to various embodiments of the disclosure. Here, the UE aggregates a first component carrier (e.g., “CC1”) **505** and second component carrier (e.g., “CC2”) **510**. In certain embodiments, both CC1 **505** and CC2 **510** are LTE carriers configured with a short TTI of different lengths, i.e., both TTI lengths are shorter than 1 ms. As depicted, one TTI of CC1 **505** overlaps with 3 TTIs of CC2 **510**. As an example, one TTI on CC1 **505** may correspond to an LTE slot (7 OFDM symbols), whereas an UL transmission on CC2

510 uses sub-slots (e.g., either two OFDM/SC-FDMA symbols or three OFDM/SC-FDMA symbols in duration). In other embodiments, the CC1 **505** and CC2 **510** are NR carriers configured with different OFDM numerologies, such that multiple NR slots on CC2 **510** overlap with one NR slot on CC1 **505**.

[0116] In the first scenario **500**, an uplink transmission on(s) PUSCH is scheduled for sTTI-1 on CC1 **505**. Here, the UE **205** multiplexes a PHR MAC-CE in the sPUSCH in sTTI-1 on CC1 **505**. The UE **205** is further scheduled with an uplink transmission on(s) PUSCH in sTTI-2 on CC2 **510**.

[0117] According to one embodiment, the UE **205** reports Power headroom information for STTI-2 on CC2 **510** within the PHR MAC-CE transmitted in sTTI-1 on CC1 **505**. Reporting the power headroom information for sTTI-2 on CC2 has the advantage that the UE **205** reports an actual PHR, i.e., PHR is reported for a scheduled TTI. The actual PHR provides information on the used MPR for the corresponding uplink transmission to the RAN node. Therefore, for cases when the UE **205** is scheduled for an uplink transmission in one of the overlapped TTIs, the UE **205** may report the actual PH for this scheduled TTI.

[0118] FIG. **6** depicts a second scenario **600** where the UE **205** aggregates serving cells (e.g., in a carrier aggregation deployment) with different TTI lengths, according to various embodiments of the disclosure. Here, the UE **205** aggregates a first component carrier (e.g., “CC1”) **605** and second component carrier (e.g., “CC2”) **610**. In certain embodiments, both CC1 **505** and CC2 **510** are LTE carriers configured with a short TTI of different lengths, such that one TTI on CC1 may correspond to an LTE slot, and three TTIs on CC2 fit within one TTI on CC1. In other embodiments, the CC1 **605** and CC2 **610** are NR carriers configured with different OFDM numerologies, such that multiple NR slots on CC2 **610** overlap with one NR slot on CC1 **605**.

[0119] Note that in the second scenario **600**, the UE is scheduled for sPUSCH on sTTI-1 of CC1 **605** as well as sTTI-1 of CC2 **610**. In the second scenario **600**, the UE **205** reports PH information for sTTI-1 of CC2 within the PHR MAC-CE transmitted on CC1. In certain embodiments, when the UE **205** is scheduled for an uplink transmission in one of the overlapped TTIs, the UE **205** reports the power headroom information for this scheduled TTI, unless there is another sTTI having its PHR due before the PHR MAC-CE is generated. Thus, in the first scenario **500** shown in FIG. **5** (e.g., sPUSCH at sTTI-2 of CC2 is scheduled with $n+8$ timing, i.e., **8** subslots before sTTI-2), the UE **205** may report PH information for sTTI-2 of CC2 within the PHR MAC-CE transmitted on CC1 when no PHR is due in sTTI-1 of CC2 (e.g., due to expiration of periodicPHR-Timer). However, if there is a PHR due in sTTI-2 of CC2 then the UE **205** may instead report PH information for sTTI-1 of CC2 within the PHR MAC-CE transmitted on CC1 and not that of sTTI-2 of CC2.

[0120] FIG. **7** depicts a third scenario **700** where a UE aggregates serving cells (e.g., in a carrier aggregation deployment) with different TTI lengths, according to various embodiments of the disclosure. Here, the UE aggregates a first component carrier (e.g., “CC1”) **705** and second component carrier (e.g., “CC2”) **710**. Note that in the third scenario **700**, the UE is scheduled for sPUSCH on sTTI-1 of CC1 **705** and also on sTTI-1, sTTI-2, and sTTI-3 of CC2 **710**.

[0121] For cases when there is only one of the overlapping TDUs on CC2 which is scheduled for an uplink transmission, the RAN node is aware for which TDU the UE **205** has reported power headroom information. However, in case several of the overlapping TDUs on CC2 are scheduled for an uplink transmission, then the RAN node and UE **205** need a common understanding for which of the overlapping TDUs the UE **205** is to report power headroom information. According to one embodiment, the UE **205** reports power headroom information for the first of the overlapping TDUs which are scheduled for an uplink transmission. One example of reporting for the first overlapping TDU is described above with reference to FIG. **2**.

[0122] Given the third scenario **700**, where all three overlapping(s) TTIs are scheduled for an uplink transmission, e.g., sPUSCH is scheduled in sTTI-1 through sTTI-3 on CC2 (e.g., via individual UL grants or via multi-TTI UL grant), the UE **205** may report power headroom information for sTTI-1 of CC2 **710** within the PHR MAC-CE transmission on CC1 **705**. In an alternative embodiment the

UE **205** reports power headroom information for the last scheduled overlapping sTTI (e.g., sTTI-3). However, since the UL grant timing, i.e., timing between UL grant and corresponding uplink transmission, needs to be also considered, reporting power headroom for the first overlapped TTI may ensure sufficient processing time to calculate the PH information.

[0123] FIG. **8** depicts a fourth scenario **800** where the UE **205** aggregates serving cells (e.g., in a carrier aggregation deployment) with different TTI lengths, according to various embodiments of the disclosure. In various embodiments, when PHR is due on more than one CC, the UE **205** may report the PH information for the earliest PHR occasion, and the pending PHR corresponding to the longer TTI may be cancelled if a virtual PHR is reported for the CC associated with the longer TTI assuming a shorter TTI length. Here, the UE aggregates a first component carrier (e.g., “CC1”) **805** and second component carrier (e.g., “CC2”) **810**. Note that in the fourth scenario **800**, the UE is scheduled for sPUSCH on sTTI-1 of CC1 **805**. During the depicted time duration, CC2 **810** includes a TTI (e.g., subframe)-based PUSCH.

[0124] In the fourth scenario **800**, assume sPUSCH at sTTI-1 of CC1 **805** is scheduled with $n+4$ timing, i.e., 4 slots before sTTI-1, and subframe-based PUSCH is scheduled with $n+3$ timing, i.e., 3 subframes before the subframe where PUSCH is transmitted). Here, the UE **205** may report a (e.g., virtual) PHR computed assuming the PHR calculation for the UL sTTI in which the PHR is transmitted (according to 3GPP agreement), and may further cancel the PHR that was due on subframe-length TTI on CC2 **810**.

[0125] FIG. **9** depicts a fifth scenario **900** where the UE **205** aggregates serving cells (e.g., in a carrier aggregation deployment) with different TTI lengths, according to various embodiments of the disclosure. Apart from the shorter TTI durations introduced for LTE Rel-15 or NR, also the minimum timing from UL grant transmission to UL PUSCH transmission has been reduced. Therefore, the different timing relations, i.e., (s) PDCCH to sPUSCH, also need to be considered when reporting power headroom information.

[0126] In the fifth scenario **900**, the UE **205** aggregates a first component carrier (e.g., “CC1”) **905** and second component carrier (e.g., “CC2”) **910**. Note that in the fifth scenario **900**, the UE is scheduled for sPUSCH on sTTI-4 of CC1 **905** and on sTTI-14 of CC2 **910**. As depicted, the UE **205** is aggregating two serving cells/CCs with different TTI lengths, where one TTI of CC1 overlaps with three TTIs of CC2 **910**. Also, the timing relations between UL grant and the corresponding UL PUSCH transmission may be different for the two aggregated serving cells.

[0127] As power headroom is calculated based on a received UL grant, i.e. estimated UL power according to the grant, when configured with different serving cells having different TTI lengths (and potentially also different timing relations, i.e. from UL grant to corresponding UL transmission), when generating the PHR MAC-CE the UE **205** may not be aware (e.g., at sTTI-0 of CC1 **905**) whether there will be some uplink transmission on the other carriers in any of the TTIs which overlap with the TTI in which PHR MAC-CE is transmitted. For example, the UE might not be fast enough to process the UL grant(s) for the overlapping TTIs on the other carriers when calculating the power headroom information. Taking the exemplary scenario depicted in FIG. **9**, when generating the PHR MAC-CE for transmission on CC1 **905** in sTTI-4, the UE is not aware of the presence of an UL grant for the overlapping sTTI(s) on CC2 **910**, i.e., sTTI-13, sTTI-14, and sTTI-15.

[0128] Therefore, the UE **205** may determine whether the power headroom value for an activated Serving Cell is based on real transmission or a reference format by considering the downlink control information which has been received until and including the physical downlink control channel (PDCCH) occasion in which the first UL grant is received since a PHR has been triggered. Here, the UE **205** considers all the UL grants having received until it receives an UL grant allocating UL resource for inclusion of PHR MAC-CE. In the example above, the UE **205** considers all UL allocations having been received until and including sTTI-0 on CC1 for determining the PHR format. Therefore, the UE **205** reports a virtual PHR for CC2 since no UL grant information is

available for the overlapping sTTIs on CC2, i.e., sTTI-13, sTTI-14, and sTTI-15. Because the UE 205 does not have UL grant information for any of the overlapping sTTIs on CC2, the UE 205 would report always a virtual PHR for CC2 regardless of how the reference PHR TTI is defined, i.e., TTI to which a reported PHR refers to.

[0129] However, considering that the UE 205 might be also configured with UL resources without dynamic scheduling, i.e., semi-persistent scheduling (SPS) or configured grants, it is still required to define a reference PHR TTI.

[0130] Assuming that the UE 205 is allocated configured grants for one or multiple of the overlapping sTTIs on CC2, the UE 205 needs to know for which of the overlapping TTI to report power headroom information. It should be noted that the UE 205 may be aware of these configured grants at sTTI-0 on CC1 and hence should consider them for PHR reporting. Here, the reference PHR TTI could be, e.g., defined—as outlined in the other embodiment—as the first of the overlapping(s) TTIs which are scheduled for an uplink transmission. Alternatively, the reference PHR TTI, i.e., TTI for which Power headroom level is computed, could be defined as the first overlapped(s) TTI.

[0131] According to another embodiment, the UE 205 reports a virtual PHR for a carrier/serving cell which has a different TTI length than the serving cell on which PHR MAC-CE is transmitted. Taken the examples depicted in the above figures, the PHR MAC-CE is transmitted on CC1 which has a TTI length of, e.g., seven OFDM symbols. As CC2 is configured 2OS-UL TTI respectively 3OS-UL TTI, the UE will report virtual for CC2, i.e., PHR calculation is performed for 7OS-UL TTI and there is no 7OS-UL allocation/transmission on CC2.

[0132] Referring back to FIG. 7, in some embodiments, the UE 205 reports a virtual power headroom corresponding to a first TTI length (e.g., slot-length TTI) on CC2 within the PHR MAC-CE transmitted in sTTI-1 on CC1, although CC2 is not configured with the first TTI length (e.g., slot-length TTI), but configured with a second TTI length (e.g., $\frac{2}{3}$ OFDM/SC-FDMA symbol duration).

[0133] To calculate the virtual PHR, the UE 205 assumes a TPC state (e.g., “f” used in the power control formula) corresponding to a third TTI length. In one example, the TPC state may be derived from one or both of the TPC states corresponding to 1 ms and $\frac{2}{3}$ symbol sTTI. For example, the third TTI can be: a) fixed/specified to 1 ms TTI, b) fixed/specified to $\frac{2}{3}$ symbol-sTTI, c) indicated/configured to one of 1 ms TTI or $\frac{2}{3}$ symbol-sTTI, or d) most recently scheduled TTI length on CC2.

[0134] In another embodiment, the UE 205 reports a power headroom for CC2 corresponding to a TTI length (e.g., 1 ms-length TTI) different than the TTI length configured for CC2 within the PHR MAC-CE transmitted in sTTI-1 on CC1. For example, referring again to FIG. 3, the UE 205 may report a virtual power headroom for CC2 within the PHR MAC-CE transmitted in sTTI-1 on CC1, wherein the TTI length that the virtual power headroom for CC2 is based on is different than the TTI length associated with the sPUSCH transmission on CC2. One example of computing a power headroom for a component carrier/serving cell based on a TTI length being different than the TTI length configured for this component carrier/serving cell is described above with reference to FIG. 3.

[0135] According to one embodiment, the UE 205 aggregating multiple serving cells reports an actual PHR for a serving cell/CC according to a configured grant, i.e., UL resources allocated without dynamic scheduling, even though there is no PUSCH transmission in the corresponding TTI on that serving cell. Here, the UE 205 transmits the PHR MAC-CE on one of the other activated serving cells on which a PUSCH transmission takes place.

[0136] For cases when UE 205 has no data available for transmission in its buffer, UE 205 may not perform an uplink transmission on PUSCH even though it has a valid resource allocation, e.g., allocated by a configured grant like SPS scheduling. Because there is no higher layer data available, the UE 205 would otherwise send a medium access control (MAC) PDU which contains only

padding bits and potentially padding buffer status report (BSR). However, from a power headroom reporting perspective, it will be still useful to send an actual PHR for the serving cell, i.e., Power headroom calculation is based on the uplink resource allocation, even when there is no PUSCH transmission, since the actual PHR provides more information to the RAN node than a virtual PHR. [0137] The actual PHR provides for example information on the used MPR for the corresponding uplink allocation, which is not provided by a virtual PHR, i.e., MPR is set to 0 dB for a virtual PHR. Furthermore, the UE **205** might know at a late point of time whether it will skip the uplink transmission or not due to changes in its transmission buffer.

[0138] FIG. **10** depicts a sixth scenario **1000** where the UE **205** aggregates serving cells (e.g., in a carrier aggregation deployment) with different TTI lengths, according to various embodiments of the disclosure. Here, the UE **205** aggregates a first component carrier (e.g., “CC1”) **1005** and second component carrier (e.g., “CC2”) **1010**. Note that in the sixth scenario **1000**, the UE **205** is scheduled for sPUSCH on sTTI-4 of CC1 and sTTI-13 and sTTI-14 of CC2.

[0139] In the sixth scenario **1000**, the UE **205** at sTTI-0 on CC1 when receiving an UL grant allocating PUSCH resources in sTTI-4 which contains a PHR MAC-CE, i.e., there was a triggered PHR pending before sTTI-0, is not aware of the presence of a UL grant received for the overlapping sTTI-13. However, the UE **205** at sTTI-0 on CC1 is aware of the configured grant for sTTI-14 on CC2. Therefore, the UE **205** may report an actual PHR for CC2 for sTTI-14 within the PHR MAC-CE contained in the PUSCH in sTTI-4 on CC1.

[0140] Because at the point of time when UE is generating the PHR MAC-CE it is only aware of the configured grant in sTTI-14, the UE reports an actual PHR for this sTTI on CC2 even though the UE **205** might later on skip the corresponding PUSCH transmission on CC2 (in sTTI-14) due to an empty buffer. It should be noted that the PHR format is different for an actual reported PHR and a virtual PHR due to the fact that the UE reports P.sub.CMAX,C for an actual PHR in addition to the PH info. Therefore, in various embodiments the UE **205** follows the determined PHR format even if, later on, the UE **205** is actually not performing a PUSCH transmission on a serving cell for which an actual PHR is reported.

[0141] According to another embodiment, the UE **205** aggregating several serving cells reports an actual PHR for a serving cell according to a dynamic UL grant, i.e., UL resource allocated by DCI, even though the UE skips the corresponding PUSCH transmission on this serving cell since there is no data available for transmission in the buffer. Similar as for the previous embodiment, reporting an actual PHR is beneficial for the scheduler since the actual PHR, i.e., power headroom information calculation computed according to the received UL grant, provides information on the MPR used for the corresponding UL allocation.

[0142] Referring again to FIG. **9**, the UE **205** may transmit a PHR MAC-CE in sTTI-4 on CC1 including a virtual PH for CC2 because the UE is not aware of any uplink allocations in the “overlapping” sTTIs on CC2 when generating the PHR MAC-CE. In the case where the UE **205** receives later in sTTI-8 on CC2 an UL grant for sTTI-14, the UE **205** may include a PHR MAC-CE in the corresponding PUSCH transmission in sTTI-14 which includes an actual PHR for CC2 (actual PHR computed based on UL grant for sTTI-14). This second PHR MAC-CE provides more detailed information to the scheduler in the RAN node, e.g., because it also provides information on the used MPR for CC2.

[0143] For cases where a “longer” sTTI overlaps with several “shorter” sTTIs and several of these overlapping(s) TTIs are scheduled for an uplink transmission, a rule (e.g., network policy) allows for unambiguous determination of which of the overlapping(s) TTIs UE reports power headroom information.

[0144] The PHR is used to give the scheduler (in RAN node) an indication whether additional resources can be scheduled without power scaling at the UE **205**, or whether resources should be reduced to avoid the power scaling. Therefore, according to another embodiment, the UE **205** reports the smallest PHR across the scheduled “overlapping” sTTIs. Referring again to FIG. **7**, the

UE **205** may report for CC2, the smallest actual PHR of the multiple (scheduled) overlapping sTTIs (e.g., that of sTTI-3 on CC2).

[0145] According to another embodiment, the PHR MAC-CE includes a field to indicate which TDU was used for the PHR computation for CC2. Then the scheduler in the RAN node unambiguously knows which UL grant is the reference for the PH computation. Including the information explicitly in the PHR MAC-CE would be also a safeguard against the loss of an UL grant, i.e., a situation where the base station scheduled sTTI-1 through sTTI-3 on CC2, but the UE **205** only detects the scheduling for sTTI-2 and sTTI-3. In that case, if there is no explicit TDU indicator included in the PHR MAC-CE, but only the rule exists that the PHR is based on the first overlapping scheduled sTTI, the UE **205** would report for sTTI-2 on CC2, but the RAN nodes would (incorrectly) interpret it as a PHR valid for sTTI-1 on CC2.

[0146] Furthermore, for the case where the UE **205** is free to choose which TDU is the reference for the reported sTTI (i.e., the UE **205** would be free to not report the minimum PHRs for that carrier), the indication of the reference TDU is used by the scheduler to correctly interpret the reported PH information and to use this information for future uplink resource allocations.

[0147] In one embodiment, on a first CC (also referred to as “CC1”) in which the PHR MAC-CE is not transmitted, if PUSCH is scheduled on this carrier (i.e., CC1) in the subframe in which the PHR is transmitted on the second CC (also referred to as “CC2”), and an sPUSCH also is scheduled on the first CC or sPUCCH is transmitted on the first CC, then the PUSCH transmission is stopped or dropped in the subframe on the first CC and sPUSCH/sPUCCH is transmitted on the first CC. Here, the PHR for this carrier (i.e., CC1) is an actual PHR for the scheduled PUSCH.

[0148] FIG. **11** depicts one embodiment of a method **1100** for receiving a paging message, according to embodiments of the disclosure. In some embodiments, the method **1100** is performed by a remote unit, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. In certain embodiments, the method **1100** may be performed by a processor executing program code, for example, a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

[0149] The method **1100** begins and the remote unit communicates **1105** with a base unit using a first uplink carrier and a second uplink carrier concurrently. Here, each carrier has a different transmission duration unit length, wherein the first uplink carrier has a longer transmission duration unit length than the second uplink carrier. In some embodiments, each of the first and second uplink carriers is associated with a different serving cell.

[0150] The method **1100** includes receiving **1110** an uplink resource allocation for a first transmission duration unit on the first uplink carrier. Here, the first transmission duration unit overlaps in time with at least two second transmission duration units on the second uplink carrier.

[0151] In certain embodiments, the remote unit also receives an uplink resource allocation for at least one of the overlapping second transmission duration units on the second uplink carrier.

[0152] The method **1100** includes determining **1115** a third transmission duration unit on the second uplink carrier. Here, the third transmission duration unit comprises at least one of the second transmission duration units.

[0153] The method **1100** includes calculating **1120** PH information for the second uplink carrier associated with the third transmission duration unit and reporting **1125** the PH information in an uplink transmission on the first transmission duration unit. The method **1100** ends. In some embodiments, the PH information is calculated according to a predefined reference format. In certain embodiments, the PH information is calculated assuming the apparatus is not scheduled to transmit a PUSCH in the third transmission duration unit.

[0154] In some embodiments, the first transmission duration unit corresponds to a slot on the first uplink carrier and the second transmission duration unit corresponds to a slot on the second uplink carrier. In certain embodiments, multiple second slots on the second uplink carrier fully overlap with the first slot. In such embodiments, calculating **1120** PH information may include calculating PH

information for a first PUSCH scheduled on the first of the multiple second slots that fully overlaps with the first slot on the first uplink carrier, wherein reporting **1125** the PH information for the second uplink carrier in an uplink transmission on the first slot comprises transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH. In certain embodiments, the first uplink carrier is configured with a first SCS and the second uplink carrier is configured with a second SCS, wherein the first SCS is smaller than the second SCS.

[0155] In some embodiments, the first transmission duration unit corresponds to a TTI of the first uplink carrier and the second transmission duration unit corresponds to a TTI of the second uplink carrier. In certain embodiments, the first uplink carrier is configured with a first TTI length and the second uplink carrier is configured with a second TTI length, wherein the first TTI length is larger than the second TTI length. In such embodiments, calculating **1120** PH information for the second uplink carrier associated with the third transmission duration unit comprises calculating PH for a TTI duration longer than the second TTI length.

[0156] In certain embodiments, the length of the third transmission duration unit is greater than the length of the first TTI. In certain embodiments, the third transmission duration unit is equal to a subframe. In certain embodiments, calculating **1120** PH information for the second uplink carrier associated with the third transmission duration unit comprises calculating PH information for a subframe containing the first TTI. In certain embodiments, the third transmission duration unit contains multiple TTIs of the second uplink carrier.

[0157] In various embodiments, the reporting **1125** PH information comprises reporting a power headroom level computed based on the uplink resource allocation. In some embodiments, the at least one of the second transmission duration units on the second uplink carrier (e.g., second TTI on the second uplink carrier) has a shortened TTI length that is less than 1 millisecond. In some embodiments, the PH information is calculated based on an uplink resource allocation received for the third transmission duration unit. In such embodiments, the uplink transmission on the third transmission duration may be either stopped or dropped.

[0158] FIG. **12** depicts one embodiment of a method **1200** for reporting power headroom information, according to embodiments of the disclosure. In various embodiments, the method **1200** is performed by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. In some embodiments, the method **1200** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

[0159] The method **1200** begins and receives **1205** a configuration for a first serving cell for PUSCH transmissions having a first SCS and a second serving cell for PUSCH transmissions having a second SCS, wherein the first SCS is smaller than the second SCS. The method **1200** includes receiving **1210** an uplink resource allocation for a first slot on the first serving cell, wherein the first slot overlaps in time with multiple second slots on the second serving cell. The method **1200** includes receiving **1215** an uplink resource allocation for at least one of the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. The method **1200** includes calculating **1220** PH information for the second serving cell for a first PUSCH scheduled on a first slot of the multiple second slots that fully overlaps with the first slot on the first serving cell. The method **1200** includes transmitting **1225** the PH information in an uplink transmission on the first slot on the first serving cell. The method **1200** ends.

[0160] FIG. **13** depicts one embodiment of a method **1300** for reporting power headroom information, according to embodiments of the disclosure. In various embodiments, the method **1300** is performed by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. In some embodiments, the method **1300** is performed by a processor, such as a microcontroller, a microprocessor, a CPU, a GPU, an auxiliary processing unit, a FPGA, or the like.

[0161] The method **1300** begins and receives **1305** a configuration for a first serving cell having a first TTI length and a second serving cell having a second TTI length, where the second TTI length is smaller than the first TTI length. The method **1300** includes receiving **1310** an uplink resource

allocation for a first TTI on the first serving cell, where the first TTI overlaps in time with multiple second TTIs on the second serving cell. The method **1300** includes calculating **1315** PH information for the second serving cell for a third TTI associated with a third TTI length that contains the first TTI on the first serving cell. The method **1300** includes transmitting **1320** the PH information in an uplink transmission on the first TTI on the first serving cell. The method **1300** ends.

[0162] Disclosed herein is a first apparatus for reporting power headroom information, according to embodiments of the disclosure. The first apparatus may be implemented by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**, described above. The first apparatus includes a transceiver and a processor. The first apparatus is configured with (i.e., the processor receives a configuration via the transceiver for) a first serving cell for PUSCH transmissions having a first SCS and a second serving cell for PUSCH transmissions having a second SCS, where the first SCS is smaller than the second SCS. The first apparatus has (i.e., the processor receives via the transceiver) an uplink resource allocation for a first slot on the first serving cell, where the first slot overlaps in time with multiple second slots on the second serving cell. Additionally, the first apparatus has (i.e., the processor receives via the transceiver) an uplink resource allocation for at least one of the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. The processor calculates PH information for the second serving cell for a first PUSCH scheduled on a first slot of the multiple second slots that fully overlaps with the first slot on the first serving cell and controls the transceiver to transmit the PH information in an uplink transmission on the first slot on the first serving cell.

[0163] In some embodiments, the processor further receives (via the transceiver) a second uplink resource allocation for at least one of the overlapping second slots on the second uplink carrier. In some embodiments, transmitting the PH information in an uplink transmission on the first slot on the first serving cell includes transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH scheduled on the first of the multiple second slots that is fully overlapped by the first slot.

[0164] In some embodiments, transmitting the PH information includes transmitting a Type 1 power headroom report. In some embodiments, transmitting the PH information includes transmitting a PHR MAC-CE in the uplink transmission in an allocated uplink allocation on the first serving cell.

[0165] In some embodiments, the first slot corresponds to a TTI of the first serving cell and the second slots correspond to TTIs of the second serving cell. In certain embodiments, the PH information for the second serving cell is calculated for TTI duration that contains multiple TTIs of the second serving cell.

[0166] In some embodiments, the first serving cell is configured with a first TTI length and the second serving cell is configured with a second TTI length, where the first TTI length is larger than the second TTI length. In certain embodiments, calculating PH information for the second serving cell includes calculating PH information for a subframe that uses the first TTI length.

[0167] In certain embodiments, calculating PH information for the second serving cell includes calculating PH for a TTI duration longer than the second TTI length. In one embodiment, the PH for the second serving cell is calculated for TTI duration that is greater than the length of the first TTI. In another embodiment, the PH for the second serving cell is calculated for TTI duration that is equal to a subframe.

[0168] In some embodiments, the PH information is calculated according to a predefined reference format. In certain embodiments, the PH information is calculated assuming the UE is not scheduled to transmit a PUSCH in the at least two second slots that fully overlaps with the first slot.

[0169] In some embodiments, the at least one of the second slots on the second serving cell has a shortened TTI length that is less than 1 millisecond. In some embodiments, the reported PH information includes a power headroom level computed based on the uplink resource allocation.

[0170] In some embodiments, the PH information is calculated based on an uplink resource allocation received for a transmission duration corresponding to multiple second slots on the second

uplink cell that are overlapped by the first slot on the first serving cell. In certain embodiments, an uplink transmission on the transmission duration is one of: stopped and dropped.

[0171] Disclosed herein is a first method for reporting power headroom information, according to embodiments of the disclosure. The first method may be performed by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**, described above. The first method includes being configured with a first serving cell for PUSCH transmissions having a first SCS and a second serving cell for PUSCH transmissions having a second SCS, where the first SCS is smaller than the second SCS. The first method includes having an uplink resource allocation for a first slot on the first serving cell, where the first slot overlaps in time with multiple second slots on the second serving cell and having an uplink resource allocation for at least one of the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. The first method includes calculating PH information for the second serving cell for a first PUSCH scheduled on a first slot of the multiple second slots that fully overlaps with the first slot on the first serving cell. The first method includes transmitting the PH information in an uplink transmission on the first slot on the first serving cell.

[0172] In some embodiments, the first method further includes receiving a second uplink resource allocation for at least one of the overlapping second slots on the second uplink carrier. In some embodiments, transmitting the PH information in an uplink transmission on the first slot on the first serving cell includes transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH scheduled on the first of the multiple second slots that is fully overlapped by the first slot.

[0173] In some embodiments, transmitting the PH information includes transmitting a Type 1 power headroom report. In some embodiments, transmitting the PH information includes transmitting a PHR MAC-CE in the uplink transmission in an allocated uplink allocation on the first serving cell.

[0174] In some embodiments, the first slot corresponds to a TTI of the first serving cell and the second slots correspond to TTIs of the second serving cell. In certain embodiments, the PH information for the second serving cell is calculated for TTI duration that contains multiple TTIs of the second serving cell.

[0175] In some embodiments, the first serving cell is configured with a first TTI length and the second serving cell is configured with a second TTI length, where the first TTI length is larger than the second TTI length. In certain embodiments, calculating PH information for the second serving cell includes calculating PH information for a subframe that uses the first TTI length.

[0176] In certain embodiments, calculating PH information for the second serving cell includes calculating PH for a TTI duration longer than the second TTI length. In one embodiment, the PH for the second serving cell is calculated for TTI duration that is greater than the length of the first TTI. In another embodiment, the PH for the second serving cell is calculated for TTI duration that is equal to a subframe.

[0177] In some embodiments, the PH information is calculated according to a predefined reference format. In certain embodiments, the PH information is calculated assuming the UE is not scheduled to transmit a PUSCH in the at least two second slots that fully overlaps with the first slot.

[0178] In some embodiments, the at least one of the second slots on the second serving cell has a shortened TTI length that is less than 1 millisecond. In some embodiments, the reported PH information includes a power headroom level computed based on the uplink resource allocation.

[0179] In some embodiments, the PH information is calculated based on an uplink resource allocation received for a transmission duration corresponding to the multiple second slots on the second serving cell that are overlapped by the first slot on the first serving cell. In certain embodiments, an uplink transmission on the transmission duration is one of: stopped and dropped.

[0180] Disclosed herein is a second apparatus for reporting PH information. In various embodiments, the second apparatus may be the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The second apparatus includes a transceiver that communicates with a base unit using a first uplink carrier and a second uplink carrier concurrently. Here, each carrier has

a different transmission duration unit length, wherein the first uplink carrier has a longer transmission duration unit length than the second uplink carrier. The second apparatus also includes a processor that receives an uplink resource allocation for a first transmission duration unit on the first uplink carrier. Here, the first transmission duration unit overlaps in time with at least two second transmission duration units on the second uplink carrier. The processor also determines a third transmission duration unit on the second uplink carrier. Here, the third transmission duration unit includes at least one of the second transmission duration units. The processor calculates PH information for the second uplink carrier associated with the third transmission duration unit and reports, via the transceiver, the PH information in an uplink transmission on the first transmission duration unit.

[0181] In some embodiments, each of the first and second uplink carriers is associated with a different serving cell. In some embodiments, the processor further receives an uplink resource allocation for at least one of the overlapping second transmission duration units on the second uplink carrier.

[0182] In some embodiments, the first transmission duration unit corresponds to a slot on the first uplink carrier and the second transmission duration unit corresponds to a slot on the second uplink carrier. In certain embodiments, multiple second slots on the second uplink carrier fully overlap with the first slot. In such embodiments, the processor calculates PH information for a first PUSCH scheduled on the first of the multiple second slots that fully overlaps with the first slot on the first uplink carrier, wherein reporting PH information for the second uplink carrier in an uplink transmission on the first slot includes transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH. In certain embodiments, the first uplink carrier is configured with a first SCS and the second uplink carrier is configured with a second SCS, wherein the first SCS is smaller than the second SCS.

[0183] In some embodiments, the first transmission duration unit corresponds to a TTI of the first uplink carrier and the second transmission duration unit corresponds to a TTI of the second uplink carrier. In certain embodiments, the first uplink carrier is configured with a first TTI length and the second uplink carrier is configured with a second TTI length, wherein the first TTI length is larger than the second TTI length. In such embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit includes calculating PH for a TTI duration longer than the second TTI length.

[0184] In certain embodiments, the length of the third transmission duration unit is greater than the length of the first TTI. In certain embodiments, the third transmission duration unit is equal to a subframe. In certain embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit includes calculating PH information for a subframe containing the first TTI. In certain embodiments, the third transmission duration unit contains multiple TTIs of the second uplink carrier.

[0185] In some embodiments, the processor calculates the PH information according to a predefined reference format. In certain embodiments, the processor calculates the PH information assuming the apparatus is not scheduled to transmit a PUSCH in the third transmission duration unit.

[0186] In various embodiments, the reported PH information includes a power headroom level computed based on the uplink resource allocation. In some embodiments, the at least one of the second transmission duration units on the second uplink carrier has a shortened TTI length that is less than 1 millisecond. In some embodiments, the PH information is calculated based on an uplink resource allocation received for the third transmission duration unit. In such embodiments, the uplink transmission on the third transmission duration may be either stopped or dropped.

[0187] Disclosed herein is a second method for reporting PH information. In various embodiments, the second method is performed by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The second method includes the UE communicating with a base unit using a first uplink carrier and a second uplink carrier concurrently. Here, each carrier has a different

transmission duration unit length, wherein the first uplink carrier has a longer transmission duration unit length than the second uplink carrier. The second method includes receiving an uplink resource allocation for a first transmission duration unit on the first uplink carrier. Here, the first transmission duration unit overlaps in time with at least two second transmission duration units on the second uplink carrier. The second method includes determining a third transmission duration unit on the second uplink carrier. Here, the third transmission duration unit includes at least one of the second transmission duration units. The second method includes calculating PH information for the second uplink carrier associated with the third transmission duration unit and reporting the PH information in an uplink transmission on the first transmission duration unit.

[0188] In some embodiments, each of the first and second uplink carriers is associated with a different serving cell. In some embodiments, the second method further includes receiving an uplink resource allocation for at least one of the overlapping second transmission duration units on the second uplink carrier.

[0189] In some embodiments, the first transmission duration unit corresponds to a slot on the first uplink carrier and the second transmission duration unit corresponds to a slot on the second uplink carrier. In certain embodiments, multiple second slots on the second uplink carrier fully overlap with the first slot. In such embodiments, the method further includes calculating PH information for a first PUSCH scheduled on the first of the multiple second slots that fully overlaps with the first slot on the first uplink carrier, wherein reporting PH information for the second uplink carrier in an uplink transmission on the first slot includes transmitting on a PUSCH a PHR that contains the PH information for the first PUSCH. In certain embodiments, the first uplink carrier is configured with a first SCS and the second uplink carrier is configured with a second SCS, wherein the first SCS is smaller than the second SCS.

[0190] In some embodiments, the first transmission duration unit corresponds to a TTI of the first uplink carrier and the second transmission duration unit corresponds to a TTI of the second uplink carrier. In certain embodiments, the first uplink carrier is configured with a first TTI length and the second uplink carrier is configured with a second TTI length, wherein the first TTI length is larger than the second TTI length. In such embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit includes calculating PH for a TTI duration longer than the second TTI length.

[0191] In certain embodiments, the length of the third transmission duration unit is greater than the length of the first TTI. In certain embodiments, the third transmission duration unit is equal to a subframe. In certain embodiments, calculating PH information for the second uplink carrier associated with the third transmission duration unit includes calculating PH information for a subframe containing the first TTI. In certain embodiments, the third transmission duration unit contains multiple TTIs of the second uplink carrier.

[0192] In some embodiments, the PH information is calculated according to a predefined reference format. In certain embodiments, the PH information is calculated assuming the apparatus is not scheduled to transmit a PUSCH in the third transmission duration unit.

[0193] In various embodiments, the reported PH information includes a power headroom level computed based on the uplink resource allocation. In some embodiments, the at least one of the second transmission duration units on the second uplink carrier has a shortened TTI length that is less than 1 millisecond. In some embodiments, the PH information is calculated based on an uplink resource allocation received for the third transmission duration unit. In such embodiments, the uplink transmission on the third transmission duration may be either stopped or dropped.

[0194] Disclosed herein is a third apparatus for reporting power headroom information. The third apparatus may be implemented by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The third apparatus includes a memory and a processor coupled to the memory, where the processor is configured to: A) receive an uplink resource allocation for a first TTI on the first serving cell, where the first TTI overlaps in time with multiple second TTIs on the

second serving cell, where the apparatus is configured with a first serving cell having a first TTI length and a second serving cell having a second TTI length, and where the second TTI length is smaller than the first TTI length; B) calculate PH information for the second serving cell for a third TTI associated with a third TTI length that contains the first TTI on the first serving cell; and C) transmit the PH information in an uplink transmission on the first TTI on the first serving cell. [0195] In some embodiments, to transmit the PH information, the processor is configured to cause the apparatus to transmit a PHR MAC-CE in the uplink transmission on the uplink resource allocation on the first serving cell. In certain embodiments, the reported PHR MAC-CE includes a power headroom information for the first serving cell computed based on the uplink resource allocation.

[0196] In some embodiments, to calculate the PH information for the second serving cell for the third TTI, the processor is configured to cause the apparatus to calculate PH information for a TTI length being longer than the second TTI length. In some embodiments, the length of the third TTI is greater than the length of the first TTI. In some embodiments, the third TTI is equal to a subframe.

[0197] In some embodiments, to calculate the PH information for the second serving cell for the third TTI, the processor is configured to cause the third apparatus to calculating PH information for a subframe containing the first TTI. In some embodiments, the third TTI contains multiple TTIs of the second serving cell.

[0198] In some embodiments, the processor calculates the PH information based on a predefined reference format. In certain embodiments, the processor calculates the PH information assuming the third apparatus is not scheduled to transmit a PUSCH in the third TTI.

[0199] In some embodiments, the first TTI length is equal to a slot. In some embodiments, the at least one of the second slots on the second serving cell has a shortened TTI length that is less than 1 millisecond.

[0200] Disclosed herein is a third method for reporting power headroom information. The third method may be performed by a UE device, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The third method includes being configured with a first serving cell having a first TTI length and a second serving cell having a second TTI length, wherein the second TTI length is smaller than the first TTI length and having an uplink resource allocation for a first TTI on the first serving cell, wherein the first TTI overlaps in time with multiple second TTIs on the second serving cell. The third includes calculating PH information for the second serving cell for a third TTI associated with a third TTI length that contains the first TTI on the first serving cell and transmitting the PH information in an uplink transmission on the first TTI on the first serving cell.

[0201] In some embodiments, transmitting the PH information includes transmitting a PHR MAC-CE in the uplink transmission on the uplink resource allocation on the first serving cell. In certain embodiments, the reported PHR MAC-CE includes power headroom information for the first serving cell computed based on the uplink resource allocation.

[0202] In some embodiments, calculating the PH information for the second serving cell for the third TTI includes calculating PH information for a TTI length being longer than the second TTI length. In some embodiments, the length of the third TTI is greater than the length of the first TTI. In some embodiments, the third TTI is equal to a subframe.

[0203] In some embodiments, calculating PH information for the second serving cell for the third TTI includes calculating PH information for a subframe containing the first TTI. In some embodiments, the third TTI contains multiple TTIs of the second serving cell.

[0204] In some embodiments, the PH information is calculated based on a predefined reference format. In certain embodiments, the PH information is calculated assuming the UE is not scheduled to transmit a PUSCH in the third TTI.

[0205] In some embodiments, the first TTI length is equal to a slot. In some embodiments, the at least one of the second slots on the second serving cell has a shortened TTI length that is less than 1 millisecond.

[0206] Disclosed herein is a fourth apparatus for reporting power headroom information. The fourth apparatus may be implemented by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The fourth apparatus includes a memory coupled with a processor, the memory including instructions executable by the processor to cause the UE to determine that the UE is configured with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and to transmit a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0207] In some embodiments, the uplink transmissions include PUSCH transmissions. In some embodiments, the PHR includes a Type 1 PHR.

[0208] In some embodiments, the instructions are further executable by the processor to cause the UE to generate a PHR MAC-CE. In such embodiments, to transmit the PHR, the instructions are executable by the processor to cause the UE to transmit the PHR MAC-CE.

[0209] In some embodiments, the instructions are executable by the processor to cause the UE to calculate power headroom information for the first PUSCH associated with the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0210] In some embodiments, the instructions are executable by the processor to cause the UE to transmit the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0211] Disclosed herein is a fourth method for reporting power headroom information. The fourth method may be performed by a UE, such as the remote unit **105**, the UE **205**, and/or the user equipment apparatus **400**. The fourth method includes determining that the UE is configured with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and transmitting a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0212] In some embodiments, the uplink transmissions include PUSCH transmissions. In some embodiments, the PHR includes a Type 1 PHR.

[0213] In some embodiments, the fourth method further includes generating a PHR medium access MAC-CE, where transmitting the PHR includes transmitting the PHR MAC-CE.

[0214] In some embodiments, the fourth method further includes calculating power headroom information for the first PUSCH associated with the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0215] In some embodiments, the fourth method further includes transmitting the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0216] Disclosed herein is a fifth apparatus for reporting power headroom information. The fifth apparatus may be implemented by a base station, such as the base unit **110**, the RAN node **210**, and/or a gNB, or eNB, as described herein. The fifth apparatus includes a memory coupled with a processor, the memory including instructions executable by the processor to cause the base station to configure a UE with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and to receive a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with

the slot associated with the first active BWP.

[0217] In some embodiments, the PHR includes a Type 1 PHR. In some embodiments, the instructions are executable by the processor to cause the base station to receive the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0218] Disclosed herein is a fifth method for reporting power headroom information. The fifth method may be performed by a base station, such as the base unit **110**, the RAN node **210**, and/or a gNB, or eNB, as described herein. The fifth method includes configuring a UE with a first SCS configuration for a first active BWP of a first serving cell, where the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell, and receiving a PHR in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, where the PHR is for a first PUSCH on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0219] In some embodiments, the PHR includes a Type 1 PHR. In some embodiments, the fifth method includes receiving the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

[0220] Embodiments may be practiced in other specific forms. The described embodiments are to be considered in all respects only as illustrative and not restrictive. The scope of the invention is, therefore, indicated by the appended claims rather than by the foregoing description. All changes which come within the meaning and range of equivalency of the claims are to be embraced within their scope.

Claims

1. A user equipment (UE) for wireless communication, the UE comprising: a processor; and a memory coupled with the processor, the memory comprising instructions executable by the processor to cause the UE to: determine that the UE is configured with a first subcarrier spacing (SCS) configuration for a first active bandwidth part (BWP) of a first serving cell, wherein the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell; and transmit a power headroom report (PHR) in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, wherein the PHR is for a first physical uplink shared channel (PUSCH) on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.
2. The UE of claim 1, wherein the uplink transmissions comprise PUSCH transmissions.
3. The UE of claim 1, wherein the PHR comprises a Type 1 PHR.
4. The UE of claim 1, wherein the instructions are further executable by the processor to cause the UE to: generate a PHR medium access control-control element (MAC-CE), wherein, to transmit the PHR, the instructions are executable by the processor to cause the UE to transmit the PHR MAC-CE.
5. The UE of claim 1, wherein the instructions are executable by the processor to cause the UE to: calculate power headroom information for the first PUSCH associated with the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.
6. The UE of claim 1, wherein the instructions are executable by the processor to cause the UE to: transmit the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.
7. A method of wireless communication at a user equipment (UE), the method comprising:

determining that the UE is configured with a first subcarrier spacing (SCS) configuration for a first active bandwidth part (BWP) of a first serving cell, wherein the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell; and transmitting a power headroom report (PHR) in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, wherein the PHR is for a first physical uplink shared channel (PUSCH) on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

8. The method of claim 7, wherein the uplink transmissions comprise PUSCH transmissions.

9. The method of claim 7, wherein the PHR comprises a Type 1 PHR.

10. The method of claim 7, further comprising: generating a PHR medium access control-control element (MAC-CE), wherein transmitting the PHR comprises transmitting the PHR MAC-CE.

11. The method of claim 7, further comprising: calculating power headroom information for the first PUSCH associated with the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

12. The method of claim 7, further comprising: transmitting the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

13. A base station for wireless communication, the base station comprising: a processor; and a memory coupled with the processor, the memory comprising instructions executable by the processor to cause the base station to: configure a user equipment (UE) with a first subcarrier spacing (SCS) configuration for a first active bandwidth part (BWP) of a first serving cell, wherein the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell; and receive a power headroom report (PHR) in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, wherein the PHR is for a first physical uplink shared channel (PUSCH) on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

14. The base station of claim 13, wherein the PHR comprises a Type 1 PHR.

15. The base station of claim 13, wherein the instructions are executable by the processor to cause the base station to: receive the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

16. A method of wireless communication at a base station, the method comprising: configuring a user equipment (UE) with a first subcarrier spacing (SCS) configuration for a first active bandwidth part (BWP) of a first serving cell, wherein the first SCS configuration is smaller than a second SCS configuration for a second active BWP of a second serving cell; and receiving a power headroom report (PHR) in a slot associated with the first active BWP that overlaps with a plurality of slots associated with the second active BWP, wherein the PHR is for a first physical uplink shared channel (PUSCH) on a first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.

17. The method of claim 16, wherein the PHR comprises a Type 1 PHR.

18. The method of claim 16, further comprising: receiving the PHR for the first PUSCH on the first slot of the plurality of slots associated with the second active BWP that fully overlaps with the slot associated with the first active BWP.
